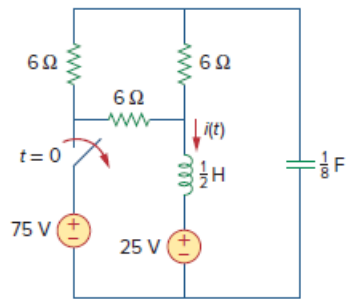


Problem

For the network in Fig. 16.55, solve for $i(t)$ for $t > 0$.

Figure 16.55:



Step-by-step solution

Step 1 of 9

Refer to Figure 16.55 in the textbook.

Consider that the switch opens at $t = 0$. Therefore, the circuit is in steady state for $t < 0$.

The capacitor acts as an open circuit and the inductor acts as a short circuit in steady state.

Modify and redraw the circuit diagram for $t < 0$ as shown in Figure 1.

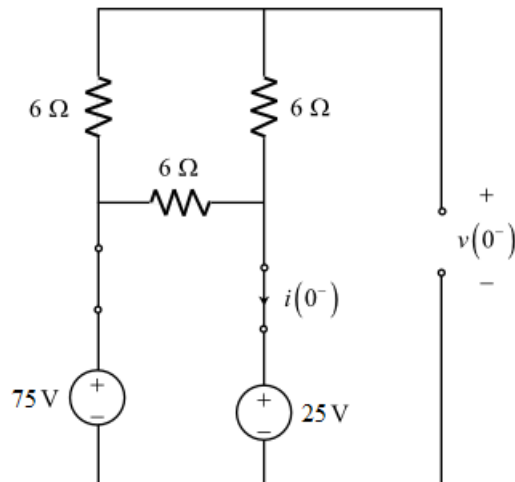


Figure 1

Step 2 of 9

Find the equivalent resistance of series combination of two $6\ \Omega$ resistors.

$$\begin{aligned} R_1 &= 6\ \Omega + 6\ \Omega \\ &= 12\ \Omega \end{aligned}$$

Simplify and redraw the circuit as shown in Figure 2.

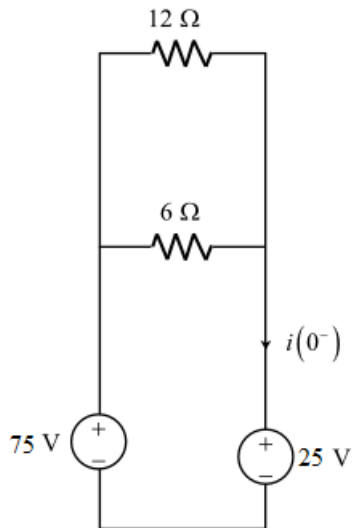


Figure 2

Step 3 of 9

Find the equivalent resistance of parallel combination of $6\ \Omega$ and $12\ \Omega$ resistors.

$$\begin{aligned} R &= 6\ \Omega \parallel 12\ \Omega \\ &= \frac{(12\ \Omega)(6\ \Omega)}{12\ \Omega + 6\ \Omega} \\ &= 4\ \Omega \end{aligned}$$

Simplify and redraw the circuit as shown in Figure 3.

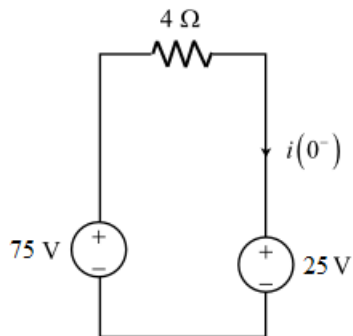


Figure 3

Step 4 of 9

Calculate the current $i(0^-)$ from the circuit in Figure 3 using Ohm's law.

$$\begin{aligned} i(0^-) &= \frac{75\ \text{V} - 25\ \text{V}}{4\ \Omega} \\ &= 12.5\ \text{A} \end{aligned}$$

Inductor does not allow sudden changes in the current flowing through it. Therefore,

$$\begin{aligned} i(0^-) &= i(0^+) \\ &= i(0) \\ &= 12.5\ \text{A} \end{aligned}$$

Step 5 of 9

Calculate the voltage $v(0^-)$ from the circuit in Figure 1.

Apply Kirchhoff's current law at node $v(0^-)$ for the circuit in Figure 1.

$$\frac{75 \text{ V} - v(0^-)}{6 \Omega} = \frac{v(0^-) - 25 \text{ V}}{6 \Omega}$$

$$75 \text{ V} - v(0^-) = v(0^-) - 25 \text{ V}$$

$$2v(0^-) = 100 \text{ V}$$

$$v(0^-) = 50 \text{ V}$$

Capacitor does not allow sudden changes in the voltage across it. Therefore,

$$\begin{aligned} v(0^-) &= v(0^+) \\ &= v(0) \\ &= 50 \text{ V} \end{aligned}$$

Step 6 of 9

Modify and redraw the circuit diagram for $t > 0$ as shown in Figure 4.

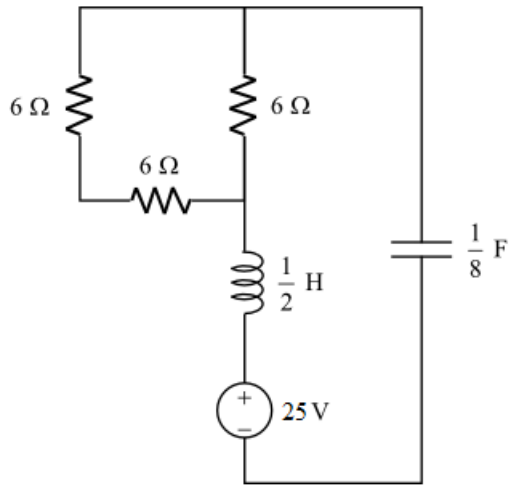


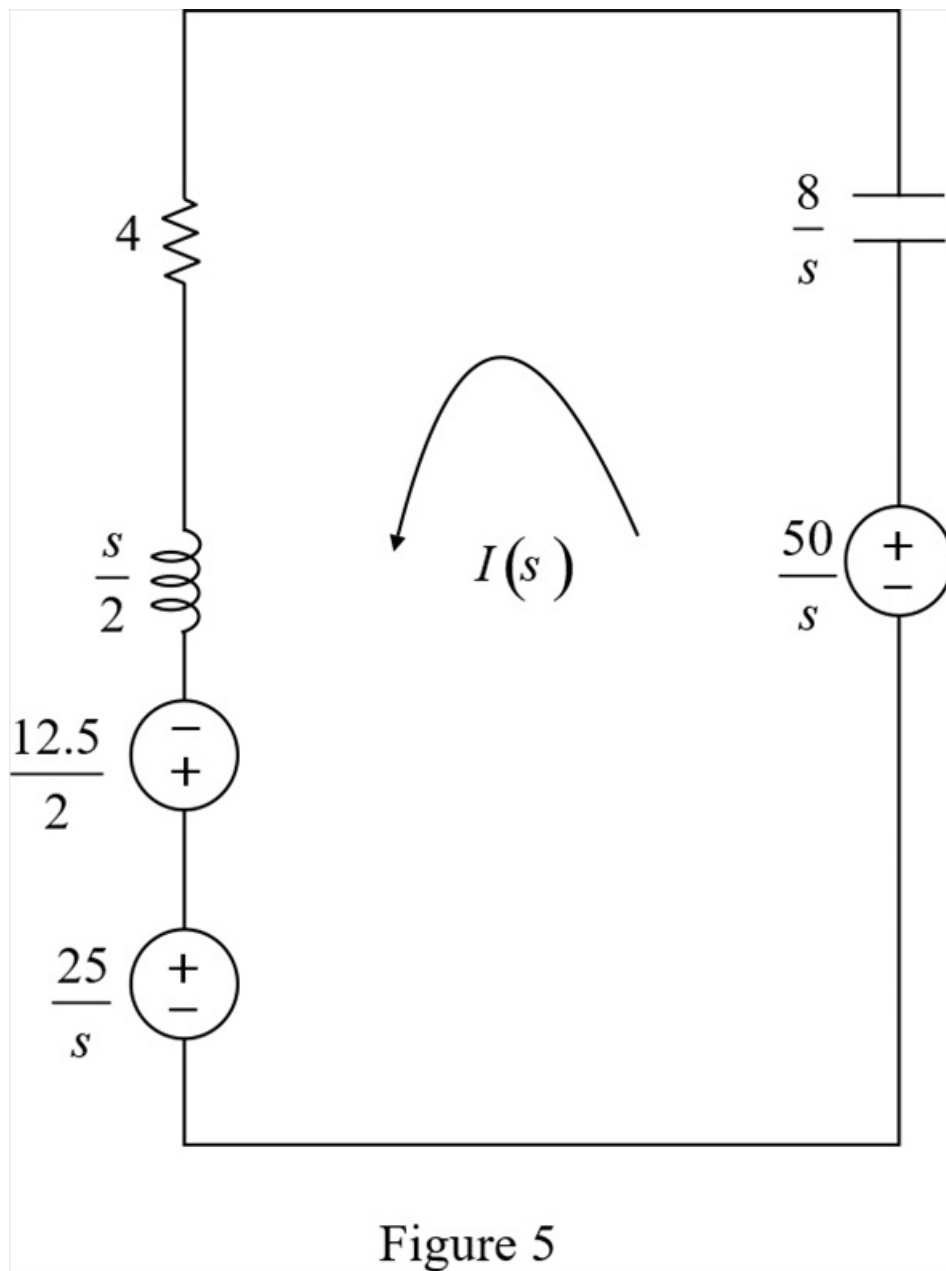
Figure 4

Step 7 of 9

Simplify the circuit in Figure 4.

$$\begin{aligned} R &= (6\ \Omega + 6\ \Omega) \parallel 6\ \Omega \\ &= 12\ \Omega \parallel 6\ \Omega \\ &= 4\ \Omega \end{aligned}$$

Represent the circuit in s-domain shown in Figure 5.



Apply Kirchhoff's voltage law to the circuit in Figure 5.

$$\frac{50}{s} - \frac{8}{s}I(s) - 4I(s) - \frac{s}{2}I(s) + \frac{12.5}{2} - \frac{25}{s} = 0$$

$$\frac{25}{s} - \frac{8}{s}I(s) - 4I(s) - \frac{s}{2}I(s) + \frac{12.5}{2} = 0$$

$$\left(\frac{8}{s} + 4 + \frac{s}{2}\right)I(s) = \frac{25 + 6.25s}{s}$$

$$\left(\frac{16 + 8s + s^2}{2s}\right)I(s) = \frac{25 + 6.25s}{s}$$

Calculate $I(s)$.

$$I(s) = \frac{25 + 6.25s}{2(s^2 + 8s + 16)}$$

$$= \frac{3.125s + 12.5}{s^2 + 8s + 16}$$

$$= \frac{3.125(s + 4)}{(s + 4)^2}$$

$$= \frac{3.125}{s + 4}$$

[Comments \(1\)](#)



Anonymous

There is a mistake here. The 2 in the denominator of I(s) belongs in the numerator, giving (50 + 12.5s)/(s^2 + 8s + 16)

Step 9 of 9

Calculate the inverse Laplace Transform to obtain $i(t)$.

$$i(t) = L^{-1}\left[\frac{3.125}{s + 4}\right]$$

$$= 3.125e^{-4t}u(t) \text{ A}$$

Hence, the value of the current is $\boxed{3.125e^{-4t}u(t) \text{ A}}$.